Deeksha Wadhwa
Software Engineer | MSc Advanced Computer Science (Merit, Newcastle University 2024–2025)
London, UK | +44 7554 959969 | deekshawadhwa78@gmail.com
https://www.linkedin.com/in/deeksha-wadhwa-984416255/ | https://github.com/Deeksha7-wa

Profile

Software Engineer with 1.5+ years' experience delivering production-grade AI/ML solutions and full-stack apps. Built Python microservices and integrated NLP & computer vision models into scalable systems, optimizing workflows and enabling engineers to make faster, smarter decisions. Passionate about high-impact, customer-facing AI tools for engineering and simulation.

Technical Skills

- **AI & ML:** Python, PyTorch, TensorFlow, GPT models, Hugging Face Transformers, NLP, Computer Vision, Deep Learning, Model Inference Pipelines, Transfer Learning, Hyperparameter Tuning
- **Backend & Deployment:** FastAPI, Flask, React, TypeScript, Docker, Kubernetes, CI/CD, AWS
- **Data & Analytics:** SQL (PostgreSQL, MySQL, SQLite), Pandas, NumPy, Scikit-learn, Feature Engineering
- **Systems & Practices:** Distributed Systems, Concurrency, Performance Tuning, Agile, Software Engineering Best Practices, Workflow Automation

Education

- **MSc Advanced Computer Science** – Newcastle University, UK | 2024–2025 | **Merit** (64.5%)
  Key Modules: Deep Learning (96%), Machine Learning, Secure Software Development, Information Security & Cryptography
- **MSc Applied Mathematics & Computing** – Devi Ahilya University, India | 2020–2022 | **Distinction** (94.7%)

Professional Experience

Software Developer – Tata Consultancy Services (TCS), India | Dec 2021 – Apr 2023

- Built Python microservices and REST APIs supporting ML workflows, improving reliability and reducing production errors by 40%.
- Integrated ML/AI components into production pipelines, automating workflows and improving system throughput for end users.
- Collaborated with cross-functional teams to deliver solutions that enabled engineers to make faster, data-driven decisions.

Visiting Faculty – AI/ML & Programming – Devi Ahilya University | Aug 2022 – Aug 2023

- Taught Python, ML, NLP, and AI concepts to 200+ students; supervised 5 real-world AI projects.

Customer Assistant – Tesco, Sunderland, UK | Apr 2025 – Jun 2025

- Improved operational efficiency by 15% through process optimization in a high-pressure environment.

Key Projects

Virtual Humanoid Female Robot (MSc Dissertation) – HTML, Python/Flask, Rasa NLP, OpenCV

- Developed a real-time conversational robot (>95% accuracy) integrating NLP and computer vision.
- Delivered a scalable, API-driven solution enabling multi-modal user interactions.

Codementor AI – FastAPI, React, SQLite, GPT-2, Vercel

- Built a full-stack ML platform with automated CI/CD deployment pipelines.
- Integrated GPT-2 for intelligent code suggestions, reducing query resolution time by 30%.

CycleGAN & Bird Classifier – PyTorch, TensorFlow

- Built distributed pipelines for image translation and bird classification, achieving >90% accuracy.

Leadership & Engagement

Student Ambassador – Newcastle University | 2024–2025

- Conducted university tours for prospective students, showcasing campus facilities and providing information about courses, programs, and student life.